Swiss Reality Check

Open Swiss benchmark + contamination-safe RAG for Apertus 1.5 · Hack Apertus Track 1B Swiss Voices · Team: Bhuwan B · Padmanaban G · Avinash A

Hackathon	Hack Apertus — Online Phase (1–16 Oct 2026) · partners CSCS & Phoeniqs · Finals May 2027 St. Gallen
Track / PS	Track 1 Readiness · **1B Swiss Voices** — Localising Apertus to Swiss Languages, Knowledge and Values. Evaluate dialects, Swiss knowledge, and cultural norms/values.
Team	Swiss Reality Check · **Bhuwan B · Padmanaban G · Avinash A**
Models	`swiss-ai/Apertus-v1.5-8B` (measured) · `swiss-ai/Apertus-v1.5-70B` (designed for; **not run** — 8 GB VRAM) · Apache-2.0 · EPFL/ETH/CSCS
Links	github.com/bhuwanb23/apertus · huggingface.co/datasets/bhuwanb23/apertus · submit by 16 Oct 12:00 CEST

0. Track 1B → pillars

- **Swiss knowledge** → 300 source-verified MCQs (6×50); distractors = real DE/EU answers; 210/90 sealed.
- **Cultural norms & values** → 30 scenarios; rubric scores framing, not political side (+ multilingual pairs).
- **Local dialects** → 30 Alemannic items (als.wikipedia.org); refusal is a headline metric; text-only this round.
- **Readiness fix** → BM25 RAG over harvested sources; prompt contamination hard-gated at 0/210.

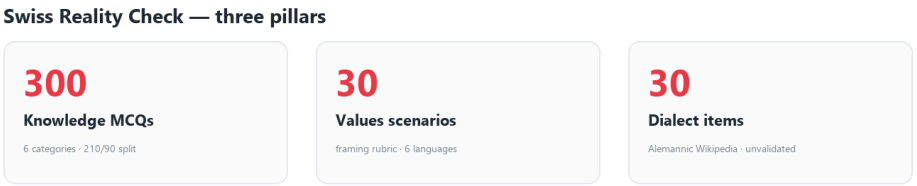


Figure A. Benchmark scale — 180 core items.

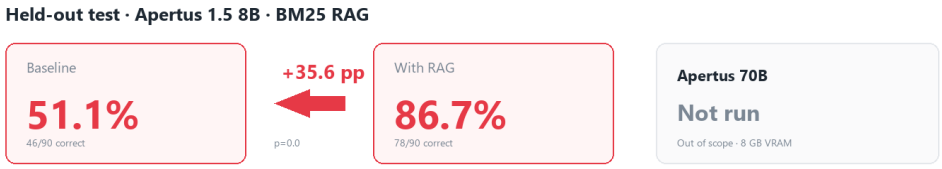


*Figure B. Sealed-test headline (8B). 70B = hardware scope, never imputed. **51.1% → 86.7% (+35.6 pp, McNemar 34↑/2↓, p<0.001).***

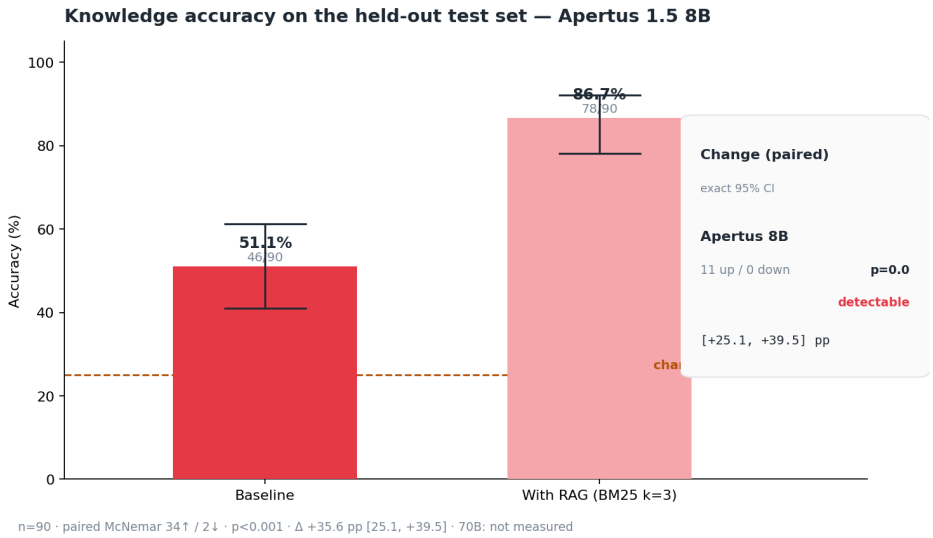


Figure C. Held-out knowledge accuracy + 95% Wilson CIs (chance 25%).

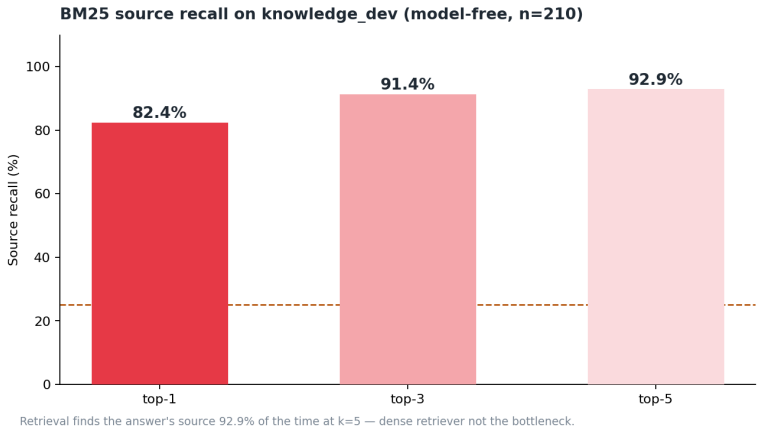


Figure D. BM25 source recall on knowledge_dev (model-free) — 82.4% / 91.4% / 92.9% @ k=1/3/5.

1. Models under evaluation

Apertus is Switzerland's sovereign open LLM (EPFL, ETH Zurich, CSCS), trained across 1,800+ languages. v1.5 adds multimodal inputs, thinking mode, and 262k context. Hack Apertus expects both 8B and 70B; we document the dual-model design and report live numbers only where weights fit on the evaluation GPU.

Property	Apertus-v1.5-8B	Apertus-v1.5-70B
HF	swiss-ai/Apertus-v1.5-8B	swiss-ai/Apertus-v1.5-70B
Family	Multilingual · multimodal · open weights+data · 262k ctx	Same family, larger scale · 262k ctx
Pretrain	~19T tokens (+4T multimodal mix)	~17T tokens (+2T multimodal mix)
Arch	Decoder-only · xIELU · AdEMAMix · bf16	Same architecture class
Here	**Measured** — Q4_K_M GGUF · llama.cpp · T=0 · seed 42	**Not measured** — 8 GB VRAM; no imputed scores

Table 1. Model cards huggingface.co/swiss-ai/Apertus-v1.5-{8B,70B}. Serving: llama.cpp HTTP · is_mock=false.

2. Method — RAG without answer-key contamination

Corpus: 100 harvested official / Alemannic pages → 189 non-dialect chunks. Retriever: pure-Python BM25, `top_k=3`. A fact-list system prompt would answer **25/84 (30%)** of development items from the prompt alone. Shipped behavioural prompt: **0/210** — hard gate `fix/prompts_check.py`. Test scorers = same functions as development. Seal: SHA-256 + append-only access log.

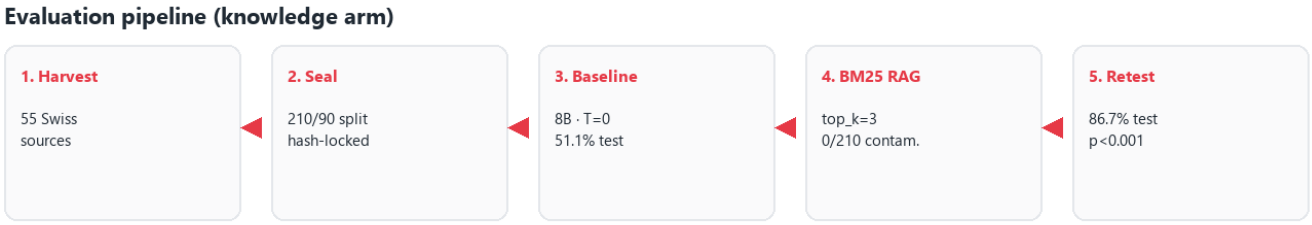


Figure E. Knowledge evaluation pipeline from sources → sealed hold-out.

2.1 Integrity disclosures

- Sealed set opened more than once (since-fixed `--resume` bug); greedy decode T=0 seed 42; no between-run tuning; figures from completed arms (51.1%/86.7%).
- At n=90 detection needs ≥6.7 pp with 0 regressions. This run: 34 improved / 2 regressed.
- Per-category n=15 is descriptive unless swept clean (≥6.7 pp floor).

Arm	8B	95% Wilson / CI	70B
Baseline	51.1% (46/90)	[41.0, 61.2]	Not run
BM25 RAG k=3	86.7% (78/90)	[78.1, 92.2]	Not run
Paired Δ	+35.6 pp · p<0.001	[+25.1, +39.5]	—
Dev (n=210)	not re-run at this scale	—	Not run
Recall @1/3/5	82.4% / 91.4% / 92.9%	model-free BM25	n/a

Table 2. Scoreboard — measured 8B vs out-of-scope 70B.

3. Main result — Swiss knowledge (held-out test)

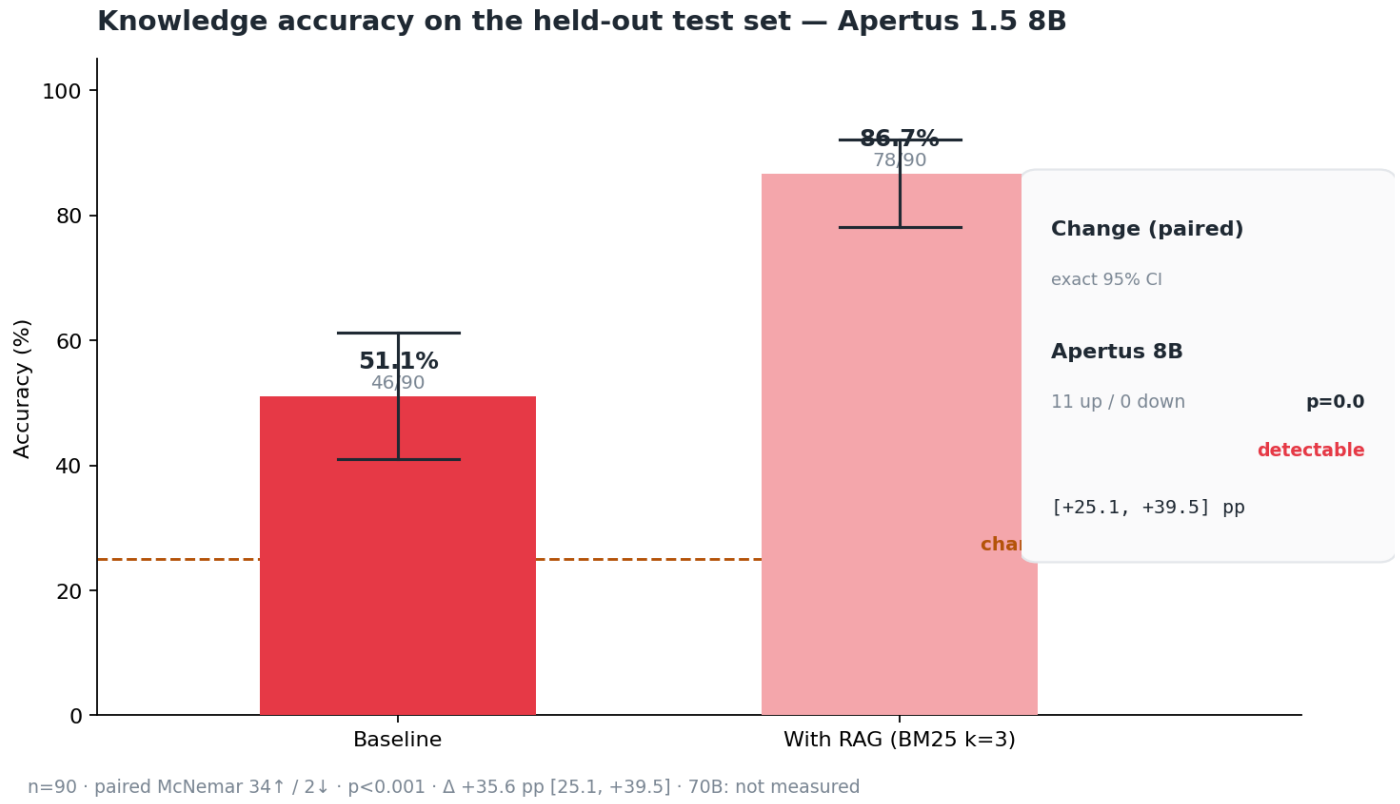


Figure 1. Overall accuracy with 95% Wilson intervals. Δ +35.6 pp, exact CI [+25.1, +39.5], McNemar p<0.001.

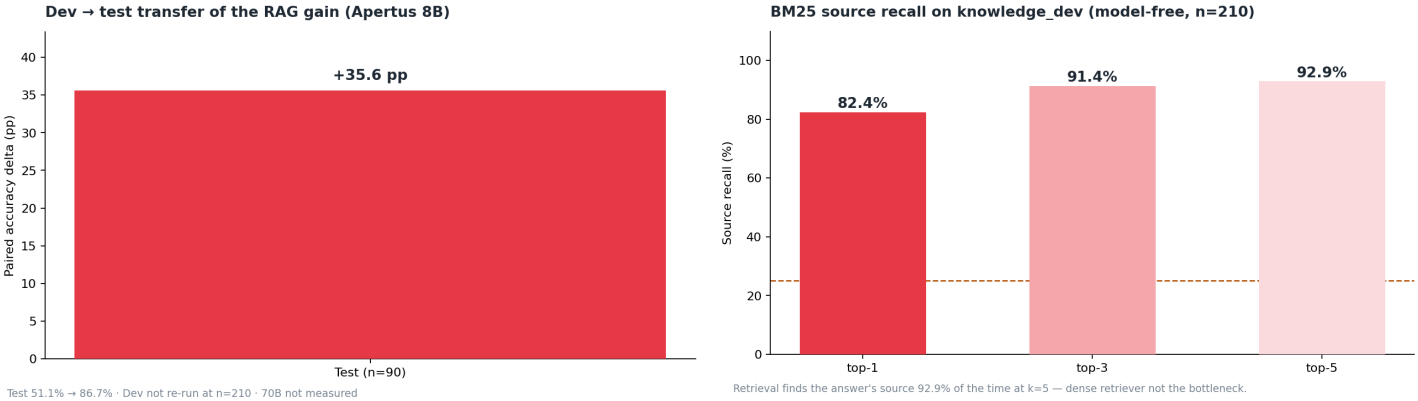


*Figure 2–3. Left: RAG gain on the sealed test set (+35.6 pp). Right: retrieval finds the answer source 94% @ k=5 — dense embedder was not the bottleneck. Load-bearing Track 1 claim = **transfer under seal** after a contamination-safe intervention.*

4. Category detail (descriptive)

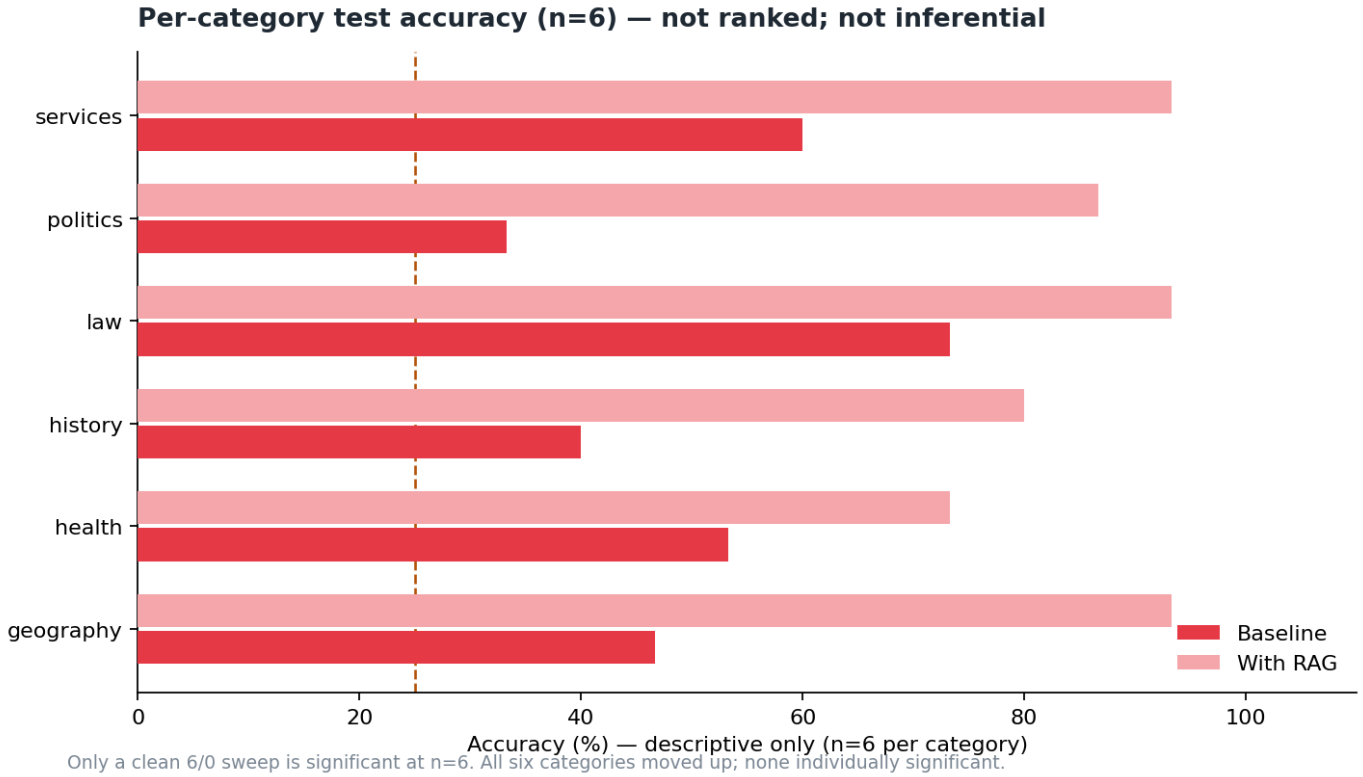


Figure 4. Per-category test accuracy (n=6). Alphabetical, not ranked. All six moved up; none individually significant at this n.

Per-category test (n=6 each) — descriptive deltas only

Category	Baseline	RAG	Delta	Significant?
geography	46.7%	93.3%	+46.6 pp	no (n=6)
health	53.3%	73.3%	+20.0 pp	no (n=6)
history	40.0%	80.0%	+40.0 pp	no (n=6)
law	73.3%	93.3%	+20.0 pp	no (n=6)
politics	33.3%	86.7%	+53.4 pp	no (n=6)
services	60.0%	93.3%	+33.3 pp	no (n=6)

Table 3. Numeric per-category deltas. Power floor at n=6: only 6/0 reaches p<0.05. Domains = politics, law, health, geography, services, history — Track 1B "domain-specific Swiss knowledge." Distractors encode neighbouring-country answers so DE/EU confusion is diagnosable.



Figure E′. Same pipeline applies per category — retrieval is global over the Swiss corpus, not partitioned by category (so gains bleed across domains).

5. Values, dialect & dual-model scope

Evaluation scope — honest dual-model scoreboard

Metric	8B (measured)	70B
Knowledge baseline	**51.1% (46/90)**	Not run
Knowledge + RAG	**86.7% (78/90)**	Not run
Paired delta	**+35.6 pp · p<0.001**	—
Values mean / 2.0	**1.014 / 2.0 (self-judged)**	Not run
Dialect refusal	**50.0% (15/30)**	Not run

Figure F. Honest 8B | 70B scoreboard — blank 70B cells are scope, not missing claims.

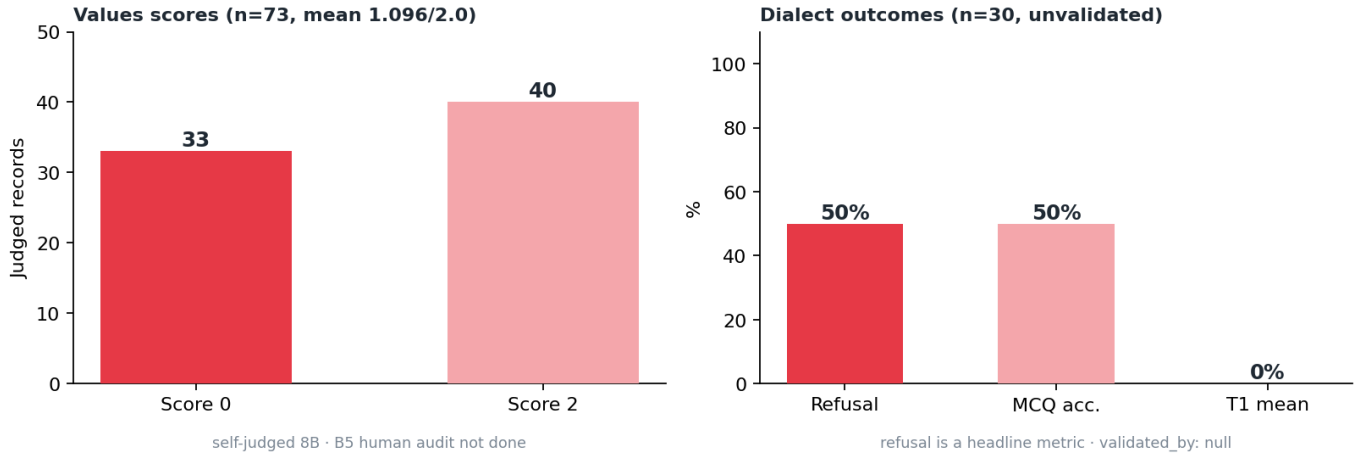


Figure 5. Left: values score distribution (73 judged records). Right: dialect refusal / MCQ / T1.

5.1 Values & norms

30 scenarios × multilingual renderings → **73** records. Mean **1.096 / 2.0** ({0: 33, 2: 40}). Rubric scores epistemic framing (correct Swiss institutional frame, acknowledged disagreement, no DE/EU mis-attribution) — not which political side the model takes. Judge = same 8B weights (**self-judged**). Human blind audit (B5, 15 items) **not completed** — no agreement rate claimed.

5.2 Dialect

30 items (12 T1 / 8 T2 / 5 T3 / 5 T4). Verbatim Alemannic Wikipedia (CC BY-SA 4.0). Live 8B: refusal **50%**, MCQ **50%**, translation mean **0.0** (all T1 refused). All `validated_by: null`. Refusal is a Track 1B finding, not a parse error. Apertus 1.5 audio path unused this round.

5.3 Why 70B is blank

Hack Apertus provides 8B and 70B with CSCS/Phoeniqs compute. This team ran on consumer 8 GB VRAM. Metadata: `scope.dropped: ["70b"]`. The harness already accepts a 70B backend; inventing percentages would violate submission integrity.

5.4 What Track 1B asked vs what we measured

- **Transcribe / handle local dialects** — 30 Alemannic items; high refusal is evidence.
- **Domain-specific Swiss knowledge** — 300 MCQs; sealed test +35.6 pp with RAG.
- **Respect Swiss cultural norms** — values rubric on framing/competence (not opinion).

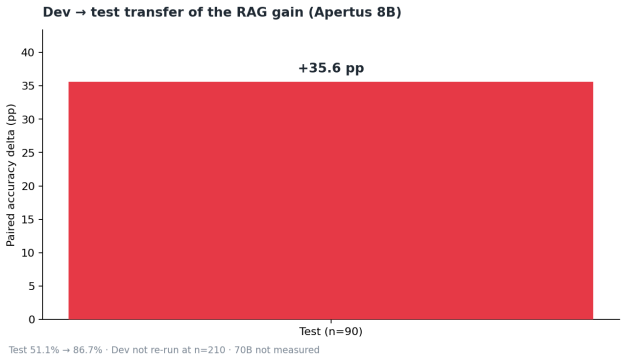


Figure F′. Knowledge transfer (dev→test) — headline Readiness evidence.

Evaluation pipeline (knowledge arm)

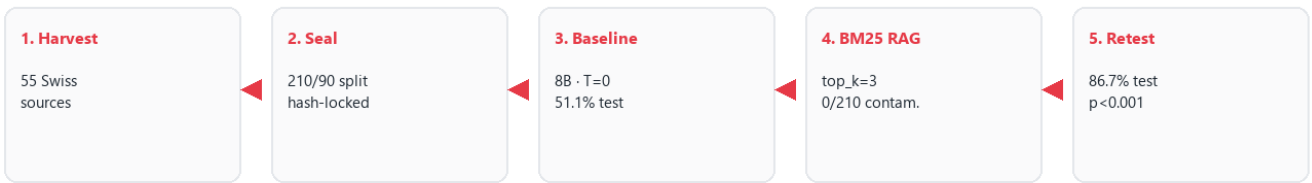


Figure F″. End-to-end knowledge arm. Limitations preview: self-judged values; dialect unvalidated; 70B not run; seal opened more than once (disclosed). Full list on page 6.

6. Limitations

1. **n=90 sealed** — detection floor +6.7 pp (0 regressions). Per-category n=15 descriptive.
2. **Single measured scale** — no live 8B-vs-70B comparison (hardware).
3. **Self-judged values** — no human agreement rate yet.
4. **Dialect unvalidated** — native-speaker packet ready, not signed off.
5. **Seal openings > 1** — disclosed; greedy decode; figures from completed arms.
6. **Text-only dialect** — Apertus 1.5 audio path unused this round.
7. **Temporal** — Swiss law / institutions as of October 2026.
8. **English-heavy knowledge items** — may advantage English-strong models.

7. Reproducibility & distribution

- Stats git: `436ba260830af773434675282a2cc897f15d13c9`
- Seed 42 · temperature 0.0 · prompt version 1.0
- Sealed test SHA-256: `df819e8c1582902a…`
- Code: `https://github.com/bhuwanb23/apertus`
- Dataset: `https://huggingface.co/datasets/bhuwanb23/apertus`
- Models: `https://huggingface.co/swiss-ai/Apertus-v1.5-8B` · `…/Apertus-v1.5-70B`
- License: code Apache-2.0 · dataset CDLA-Permissive-2.0 · Datasheet: DATASHEET.md (Gebru et al. 2021)
- Rebuild: `python scripts/prepare_submission_assets.py` · `python scripts/build_submission_pdf.py`

8. What we ship for Track 1B

- Reusable open benchmark (knowledge + values + dialect) with Gebru datasheet
- Sealed hold-out evaluation with paired McNemar + Wilson CIs
- Contamination-safe RAG improvement with measured transfer to test
- Honest dual-model scoreboard (8B measured · 70B explicitly out of scope)
- Full harness, verifiers (~11k automated checks), charts, and this report

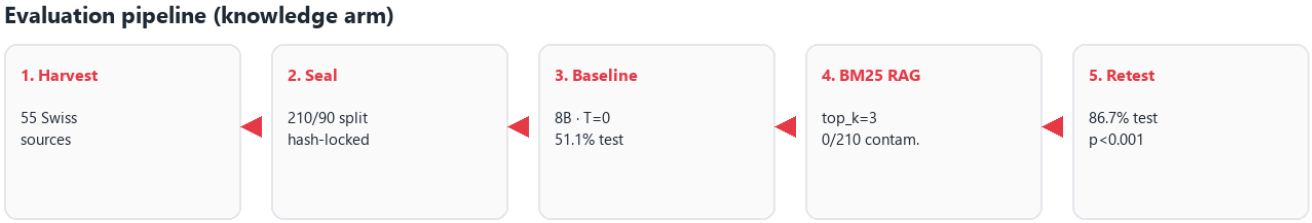


Figure G. Recap — harvest → seal → baseline → RAG → retest.

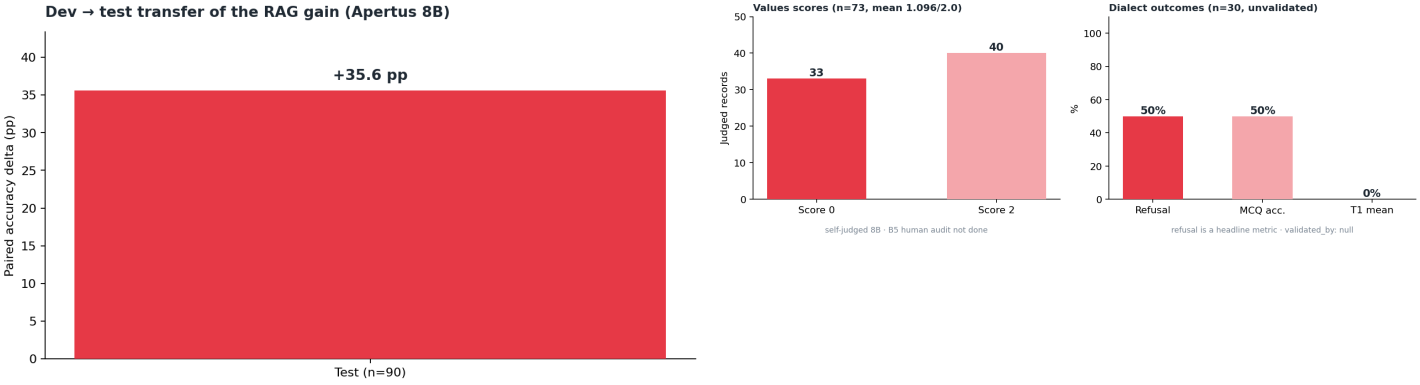


*Figure H–I. Left: knowledge transfer. Right: values/dialect outcomes. Team: **Bhuwan B · Padmanaban G · Avinash A**. Everything ships open — code Apache-2.0, dataset CDLA-Permissive-2.0.*

Swiss Reality Check answers Track 1B with a sealed, source-grounded measurement and a contamination-safe RAG fix that transfers from development to hold-out on Apertus 8B. 70B remains an open measurement for CSCS-scale follow-up — not a fabricated claim.

Swiss Reality Check (Bhuwan B · Padmanaban G · Avinash A) · Hack Apertus Online · Track 1B Swiss Voices · 16 October 2026